

Backend Networking Cheat Sheet

1. Big Picture

Client -> HTTP Request -> IP + Port -> Operating System -> Node.js/Express -> Routes -> Business Logic -> Database -> HTTP Response -> Client

2. IP Address

Identifies a computer on a network. Think: House address.

3. Port

Identifies the application listening on the computer. Think: Room number in a hotel. Example: localhost:8000 means your computer, application on port 8000.

4. HTTP Request

Made of Request Line, Headers and optionally a Body.

5. Headers

Metadata/instructions (Content-Type, Authorization, Accept). Think: Instructions on the envelope.

6. Body / Payload

Actual data being sent (often JSON). Think: The letter inside the envelope.

7. GET

Retrieve data. Should not modify server state.

8. POST

Send data to the server, usually with a request body.

9. Express

app.listen(8000) tells Node.js to listen on port 8000.

10. URL

http://localhost:8000/api/movies?year=2020 = protocol + host + port + route + query parameters.

11. Common Ports

80 HTTP, 443 HTTPS, 3000 React, 5432 PostgreSQL, 6379 Redis, 8000 Backend, 27017 MongoDB.

12. Mental Model

IP = Hotel address. Port = Room number. Application = Person inside the room. Headers = Envelope instructions. Body = Letter.